Innovation Report: Dark Mode Implementation

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Course: Mobile Applications Development Project – Movie App

1. Overview of Innovation

The primary innovation in this project is the implementation of a **Dynamic Theme Switching System (Dark Mode)**.

This feature allows users to toggle between a standard light interface and a high-contrast dark interface at the touch of a button.

While aesthetically pleasing, this feature was technically implemented to enhance user experience and accessibility.

2. Technical Implementation

The innovation was built using a combination of:

- CSS Custom Properties (Variables),
- Angular Property Binding
- DOM Manipulation.

- **CSS Variables**: Rather than hard-coding colors, I defined a global theme system within variables.css. By targeting a specific .dark class, I was able to override the Ionic color stack ($--ion-background-color$, $--ion-text-color$) using the !important declaration to ensure theme consistency across all components.
- **State Management**: Within home.page.ts, I implemented a boolean flag (isDarkMode) to track the application state.
- **DOM Integration**: I used the document.body.classList.toggle method. This approach ensures that the theme is applied globally to the entire application body, rather than just a single page, providing a seamless transition between the Home, Movie Details, and Person Details pages.

3. Design Decisions & Creative Approach

A key design decision was to use a **"Midnight Grey"** ($#121212$) rather than pure black. This reduces "black smearing" on OLED screens and improves readability by reducing the harsh contrast of pure white text on pure black backgrounds.

I also chose to place the toggle in the header as a **Moon Icon**. This provides a clear, intuitive "call to action" that users recognize immediately, fulfilling the "Innovation" requirement of making the app feel like a modern, commercially viable product.

4. Value Added

- **Accessibility**: Provides a better viewing experience for users with light sensitivity or those using the app in low-light environments (e.g., in a cinema or at night).
- **Battery Efficiency**: For users on mobile devices with OLED screens, the dark theme significantly reduces power consumption.
- **Technical Robustness**: The implementation demonstrates an understanding of the Angular lifecycle and CSS specificity, going beyond the basic project requirements to show technical curiosity.